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HumanOS

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TABLE OF CONTENTS

Overview	2
Core Features	2
User Interface	5
How to Use	6

OVERVIEW

HumanOS was made to help students and workers visualize their focus and have suggestions to keep them on track. It was made with the idea of being private, non-invasive, and helpful in mind.

It contains many features that help track, process, and display your focus as well as give you tips on how to improve it. The goal is to remain as focused on the task as possible.

Features include:

- Accurate System Tracking
- Simple Cognitive Model
- Predictive Decision Engine
- Non-Invasive Notifications
- Sleek UI

FEATURES

SYSTEM TRACKING

The system tracking measures the input of the user through keyboard, mouse, window switching, and idle time to then collect and pipe to the cognitive model to be processed and determine how focused the user is.

It tracks data on a per-minute basis and resets the next to determine how focused the user is per minute.

```
lskdjflksdajfkjaslkfjlksdjllsadkjflksjlwindow switch detected: 1
window switch detected: 2
window switch detected: 3
window switch detected: 4
Initial System Time: Some(Instant { tv_sec: 184695, tv_nsec: 398094725 })
Key Count: 47
Backspace Count: 0
Last Mouse Move: Some(Instant { tv_sec: 184705, tv_nsec: 108923500 })
Last Wheel Scroll: None
Last Activity: Some(Instant { tv_sec: 184705, tv_nsec: 108923500 })
Window Switch Count: 4

Initial System Time: Some(Instant { tv_sec: 184695, tv_nsec: 398094725 })
Key Count: 47
Backspace Count: 0
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Last Wheel Scroll: None
Last Activity: Some(Instant { tv_sec: 184705, tv_nsec: 108923500 })
Window Switch Count: 4
```
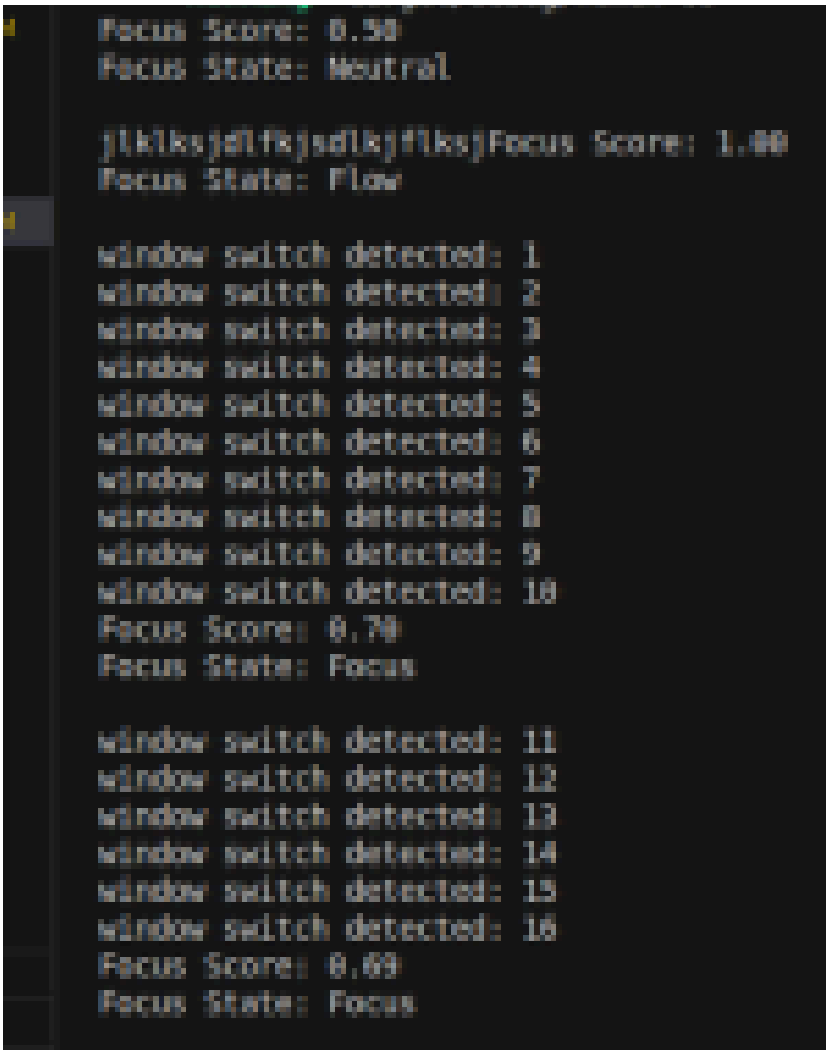

COGNITIVE MODEL

The cognitive model turns the mess of inputs into a score to determine focus. This score is then split into one of five states to further classify focus into five simple states.

In addition, the scores are saved and later plotted on a graph for the user to see focus over time.



DECISION ENGINE

The decision engine runs when the user is slipping into less focus. It will run when it detects focus going down, not just low, to ensure that the user is working with maximum efficiency. The decision engine examines system input and the overall focus score to determine what action should be taken to keep the user focused.

There are three possible actions that can be made currently: closing distracting tabs, placing notifications on do-not-disturb, and having the user take a break.

INTERVENTION

The intervention module takes action when signaled by the decision engine to do so. It executes the action decided upon by the decision action in the way each suggestion is to be handled. All intervention measures start with a notification for the user confirming that it is okay to take that action.

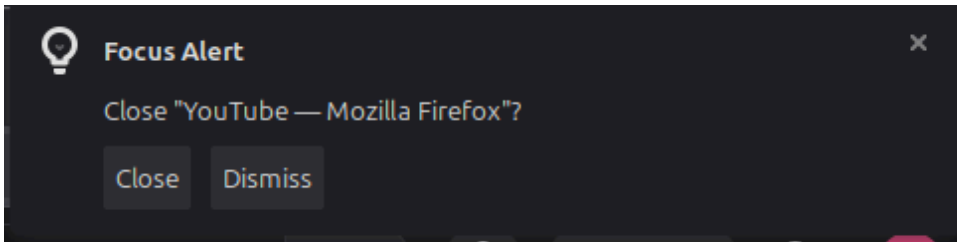
If the action is to close tabs, the program finds the distracting tabs of the user asks before closing it. At the moment, only Steam and YouTube are classified as distracting.

If the action is to put notifications on do-not-disturb, the user is asked if they want do-not-disturb enabled and, if yes, it is enabled for 10 minutes.

If the action is to ask the user if they would like a break, the user is asked and, if accepted, a timer is set for a 10 minute break.

NOTIFICATIONS

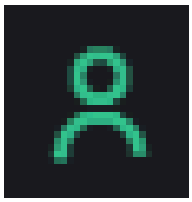
The notifications module sends notifications and receives user answer for the intervention module to use. It is meant to keep the user in the loop, not be annoying, and be as simple and quick to read as possible as to not be distracting.



USER INTERFACE

TRAY ICON

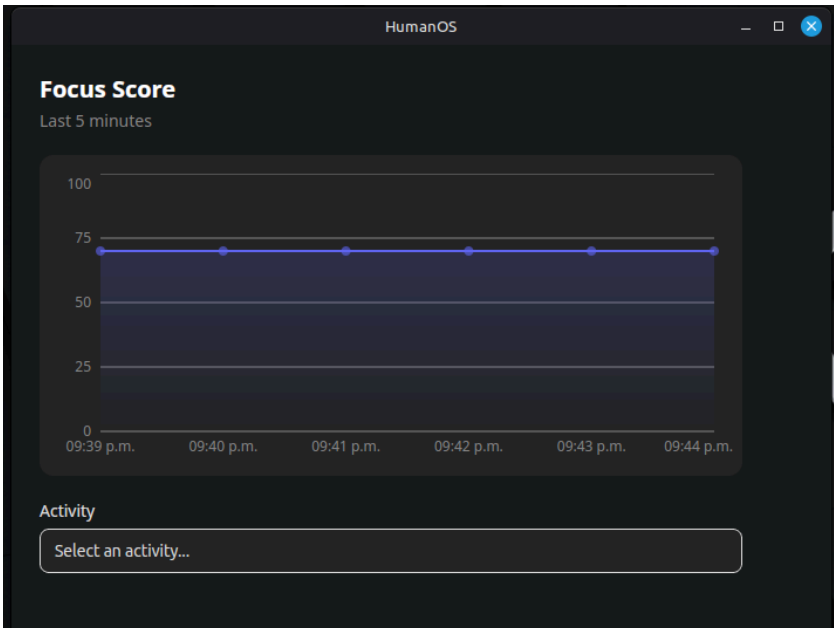
The tray icon is the main UI for this system. It changes colors from a range from red, to orange, to yellow, to green, to blue, going from fatigued to flow. It is meant to be an assessment of the focus at a glance. You can also click on the icon to see the menu with the focus score percentage and the quit button.



APP INTERFACE

The system comes with a sleek application interface to show focus score history. Right now, it only shows the last five minutes but I plan to expand it to a full day.

In addition, it has a menu where I plan to implement a target objective feature to make filtering distracting tabs more targeted towards those not related to the target objective.



HOW TO USE

SOURCE CODE

```
- Download from latest release
- Open terminal
- type `cd Downloads`
- type `./human-os` to run
```

.DEB PACKAGE

```
- Download from latest release
- Open terminal
- type `cd Downloads`
- type `sudo apt install ./human-os_0.3.0_amd64.deb`
- Run from terminal with `human-os` or use app menu
```

.APPIMAGE

```
- Download from latest release
- Open terminal
- type `cd Downloads`
- type `chmod +x human-os_0.3.0_amd64.AppImage `
- type `./human-os_0.3.0_amd64.AppImage`
```